



Assignment 01: Git, Gitea, GitHub, Git LFS and GitHub Pages

Cloud Computing Assignment

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Links:

- CC repository: <https://github.com/ummeammara60/CC>
- GitHub Assignment01: <https://github.com/ummeammara60/Assignment01>
- Portfolio repository: <https://github.com/ummeammara60/ummeammara60.github.io>
- Live site: <https://ummeammara60.github.io/>

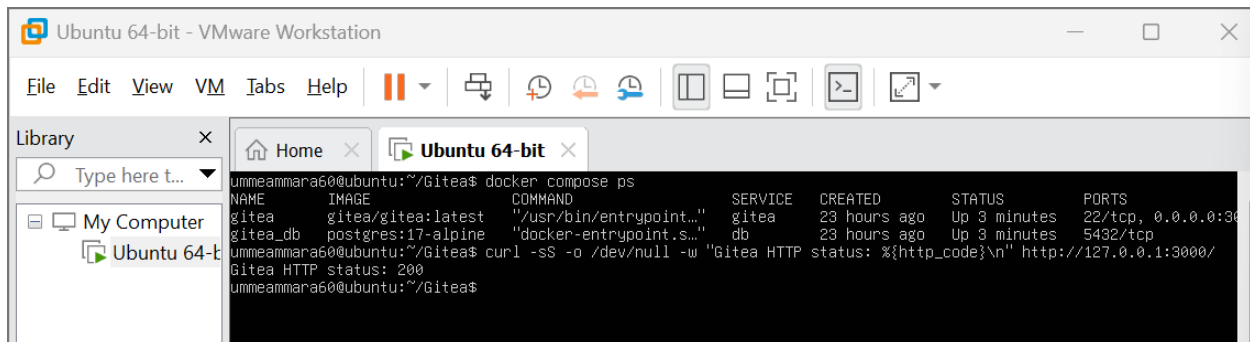
Task 1: Install Gitea and Push a Repository from the Ubuntu Server

Step 1.1: Install and run Gitea

Docker, Docker Compose and Git were installed on the Ubuntu server and Gitea with PostgreSQL was started using the instructor compose.yaml without changes.

```
sudo apt update -y
sudo apt install -y docker.io curl git-lfs
sudo mkdir -p /usr/local/lib/docker/cli-plugins
sudo curl -SL
https://github.com/docker/compose/releases/latest/download/docker-compose-
linux-x86_64 -o /usr/local/lib/docker/cli-plugins/docker-compose
sudo chmod +x /usr/local/lib/docker/cli-plugins/docker-compose
sudo systemctl start docker
sudo usermod -aG docker $USER
exit
# log in again, then:
git --version
docker --version
docker compose version
cd ~
git clone https://github.com/WaqasSaleem97/Gitea.git
cd Gitea
docker compose up -d
docker compose ps
curl -sS -o /dev/null -w "Gitea HTTP status: %{http_code}\n"
http://127.0.0.1:3000/
ip addr
```

Gitea was opened at <http://192.168.244.130:3000/> and installed with PostgreSQL (host db:5432, user/password/database gitea). Server domain 192.168.244.130, base URL <http://192.168.244.130:3000/>. An administrator account was created.



Step 1.2: Create and push the initial repository

An empty public repository Assignment01 was created in Gitea (no README, .gitignore or license). A local repository was created on the server and pushed using a personal access token (ubuntu-server-token, write:repository) entered at the password prompt.

```
cd ~
mkdir Assignment01
cd Assignment01
git init
```

```
git branch -M main
echo "# Assignment01" > README.md
echo "Name: YOUR FULL NAME" >> README.md
echo "Registration No: YOUR REGISTRATION NUMBER" >> README.md
git add README.md
git commit -m "Add student information"
git remote add gitea http://<gitea-username>@127.0.0.1:3000/<gitea-username>/Assignment01.git
git remote -v
git push -u gitea main
# paste the token at the Password prompt (nothing is shown)
```

```
ummeammara60@ubuntu:~/Assignment01$ git remote -v
gitea      http://ummeammara60@127.0.0.1:3000/ummeammara60/Assignment01.git (fetch)
gitea      http://ummeammara60@127.0.0.1:3000/ummeammara60/Assignment01.git (push)
ummeammara60@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://ummeammara60@127.0.0.1:3000':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 258 bytes | 258.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To http://127.0.0.1:3000/ummeammara60/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'gitea/main'.
ummeammara60@ubuntu:~/Assignment01$
```

Step 1.3: Verify the Gitea repository

The screenshot displays the Gitea web interface for a repository named 'Assignment01' owned by 'ummeammara60'. The interface includes a navigation bar with 'Issues', 'Pull Requests', 'Milestones', and 'Explore'. Below the repository name, there are tabs for 'Code', 'Issues', 'Pull Requests', 'Actions', 'Packages', 'Projects', 'Releases', 'Wiki', and 'Activity'. The 'Code' tab is active, showing a list of commits. The first commit is by 'ummeammara60' with hash 'cc0fe9912f', titled 'Add student information', and dated '6 minutes ago'. Below the commit list, the README content is displayed, showing 'Assignment01' and 'Name: Umme Ammara Chaudhary'. On the right side, there is a 'Description' section with 'No description provided' and a sidebar with 'Manage Topics', 'Readme', '24 KiB', '0 Stars', '1 Watchers', and '0 Forks'. The footer shows 'Powered by Gitea Version: 28.0.0 Page: 534ms Template: 129ms' and navigation links for 'Auto', 'English', 'Licenses', and 'API'.

Task 2: Push the Same Repository to GitHub

Step 2.1: Add the GitHub remote

An empty public GitHub repository Assignment01 was created. GitHub was added as the second remote to the same local repository.

```
cd ~/Assignment01
git remote add github https://github.com/ummeammara60/Assignment01.git
git remote -v
```

```
ummeammara60@ubuntu:~/Assignment01$ git remote -v
gitea    http://ummeammara60@127.0.0.1:3000/ummeammara60/Assignment01.git (fetch)
gitea    http://ummeammara60@127.0.0.1:3000/ummeammara60/Assignment01.git (push)
github   https://github.com/ummeammara60/Assignment01.git (fetch)
github   https://github.com/ummeammara60/Assignment01.git (push)
ummeammara60@ubuntu:~/Assignment01$
```

Step 2.2: Push to GitHub

```
git push -u github main
# username: ummeammara60 password: GitHub personal access token (hidden)
```

```
ummeammara60@ubuntu:~/Assignment01$ git push -u github main --force
Username for 'https://github.com': ummeammara60
Password for 'https://ummeammara60@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 258 bytes | 129.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/ummeammara60/Assignment01.git
 + b4a9f60...cc0fe99 main -> main (forced update)
branch 'main' set up to track 'github/main'.
ummeammara60@ubuntu:~/Assignment01$
```

Step 2.3: Verify the GitHub repository

https://github.com/ummeammara60/Assignment01

Chat

ummeammara60 / Assignment01

Type / to search

+ -

🔍

📄

🔗

🛡️

⚙️

<> Code

🕒 Issues

🔗 Pull requests

🔗 Agents

🕒 Actions

📁 Projects

📖 Wiki

🛡️ Security and quality

📊 Insights

⚙️ Settings

Assignment01

Public

📌 Pin

👁 Watch 0

🍴 Fork 0

★ Star 0

main 1 Branch 0 Tags

Go to file

+

<> Code

ummeammara60 Add student information cc0fe99 · 26 minutes ago 1 Commit

README.md Add student information 26 minutes ago

README

Assignment01

Name: Umme Ammara Chaudhary

About

Assignment01

📖 Readme

🔗 Activity

★ 0 stars

👁 0 watching

🍴 0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Contributors 1

Task 3: Track Three Large Files with Git LFS

Step 3.1: Configure Git LFS

```
cd ~/Assignment01
git lfs install
git lfs version
git lfs track "*.bin"
cat .gitattributes
```

```
ummeammara60@ubuntu:~/Assignment01$ git lfs install
Updated Git hooks.
Git LFS initialized.
ummeammara60@ubuntu:~/Assignment01$ git lfs version
git-lfs/3.7.1 (GitHub; linux amd64; go 1.25.0)
ummeammara60@ubuntu:~/Assignment01$ git lfs track "*.bin"
Tracking "*.bin"
ummeammara60@ubuntu:~/Assignment01$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
ummeammara60@ubuntu:~/Assignment01$
```

Step 3.2: Create and verify three LFS files

```
truncate -s 101M large-file-1.bin
truncate -s 102M large-file-2.bin
truncate -s 103M large-file-3.bin
git add .gitattributes large-file-1.bin large-file-2.bin large-file-3.bin
git lfs ls-files --size
```

```
ummeammara60@ubuntu:~/Assignment01$ git lfs ls-files --size
f1753a034b * large-file-1.bin (106 MB)
2338a970c0 * large-file-2.bin (107 MB)
cb91d5908e * large-file-3.bin (108 MB)
ummeammara60@ubuntu:~/Assignment01$
```

Step 3.3: Commit and push the LFS files

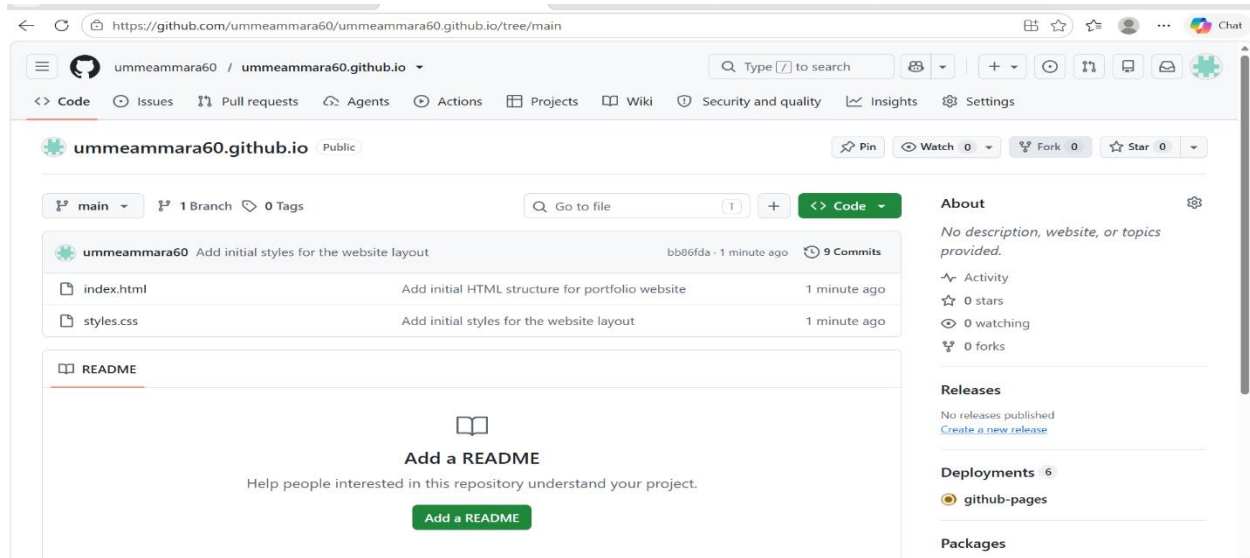
```
git commit -m "Add three files using Git LFS"
git push github main
```

```
ummeammara60@ubuntu:~/Assignment01$ git push github main
Username for 'https://github.com': ummeammara60
Password for 'https://ummeammara60@github.com':
Uploading LFS objects: 100% (3/3), 366 MB | 505 KB/s, done.
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 790 bytes | 197.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/ummeammara60/Assignment01.git
   cc0fe99..78976ee  main -> main
ummeammara60@ubuntu:~/Assignment01$ _
```

Task 4: Create a Portfolio or CV with GitHub Pages

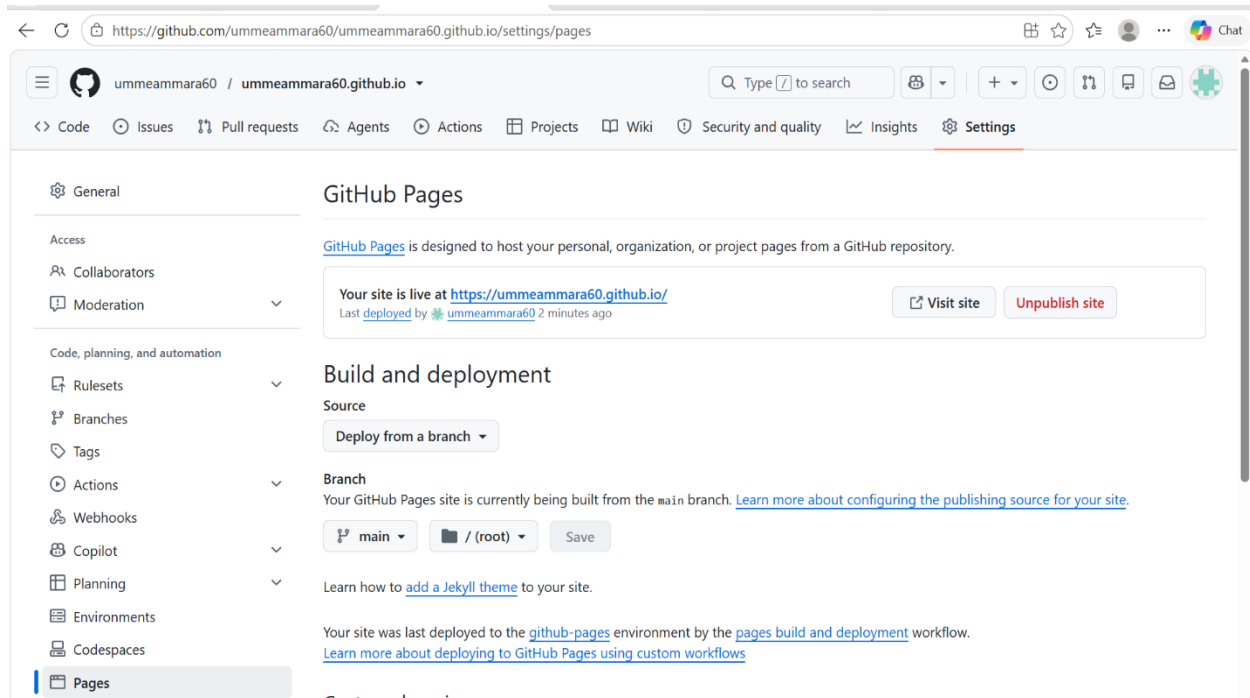
Step 4.1: Portfolio repository

A public repository named `ummeammara60.github.io` was created containing `index.html` and `styles.css` with the sections About Me, Education, Skills and Projects, pushed to the main branch.

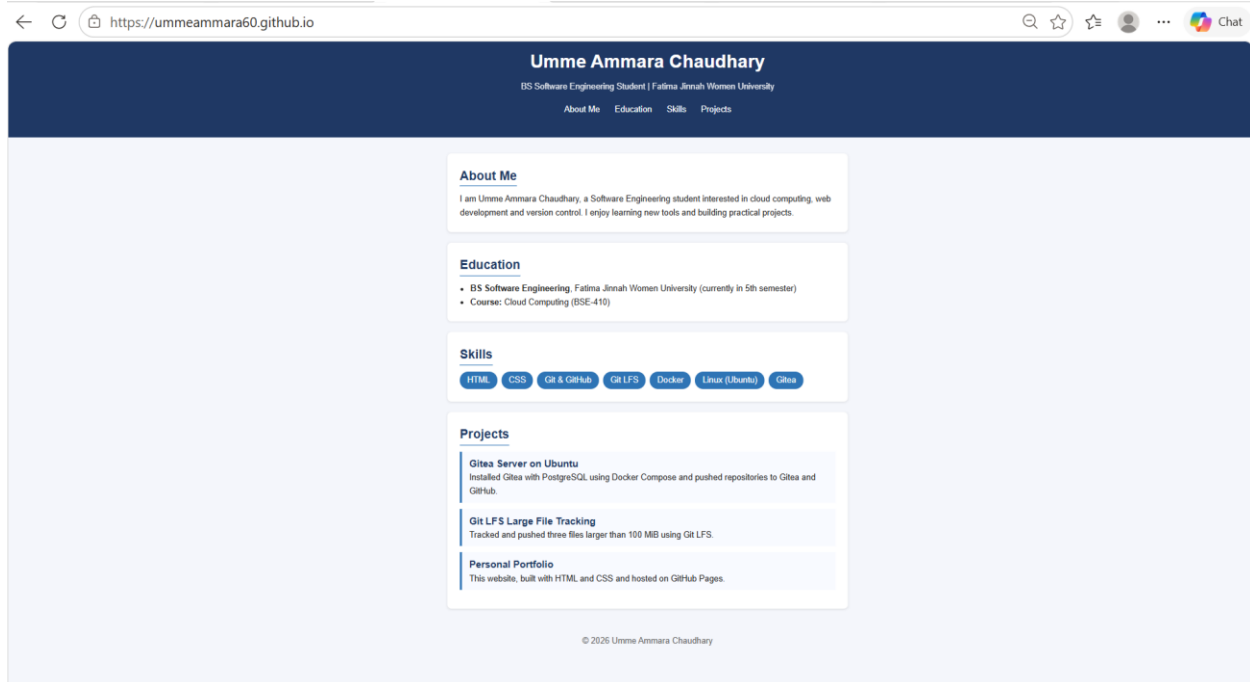


Step 4.2: Publish with GitHub Pages

Settings > Pages > Build and deployment > Deploy from a branch > main and / (root) > Save.



Step 4.3: Verify the live portfolio



Conclusion

Gitea was installed on the Ubuntu server with Docker Compose, a repository was pushed to both Gitea and GitHub using two remotes without exposing credentials, three files larger than 100 MiB were tracked with Git LFS, and a portfolio was published with GitHub Pages.